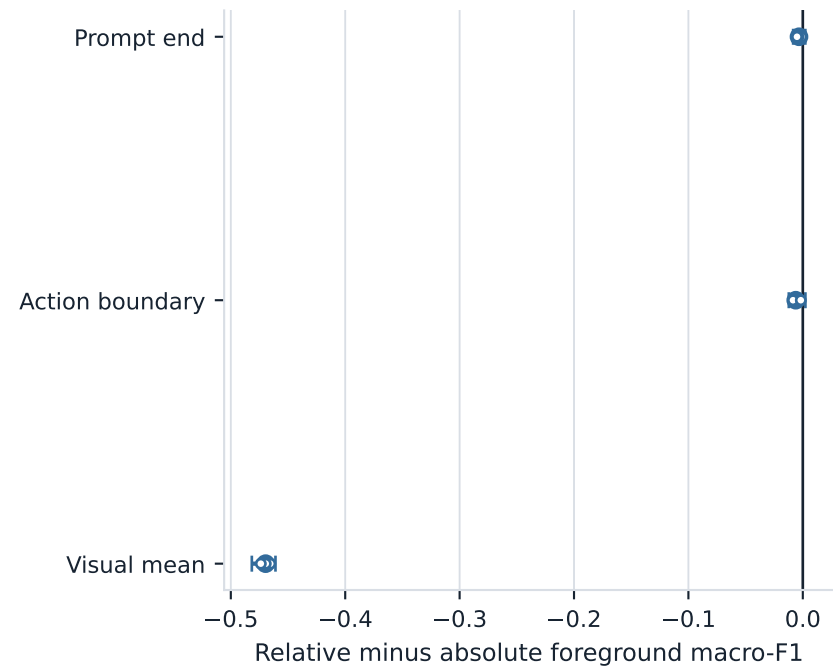


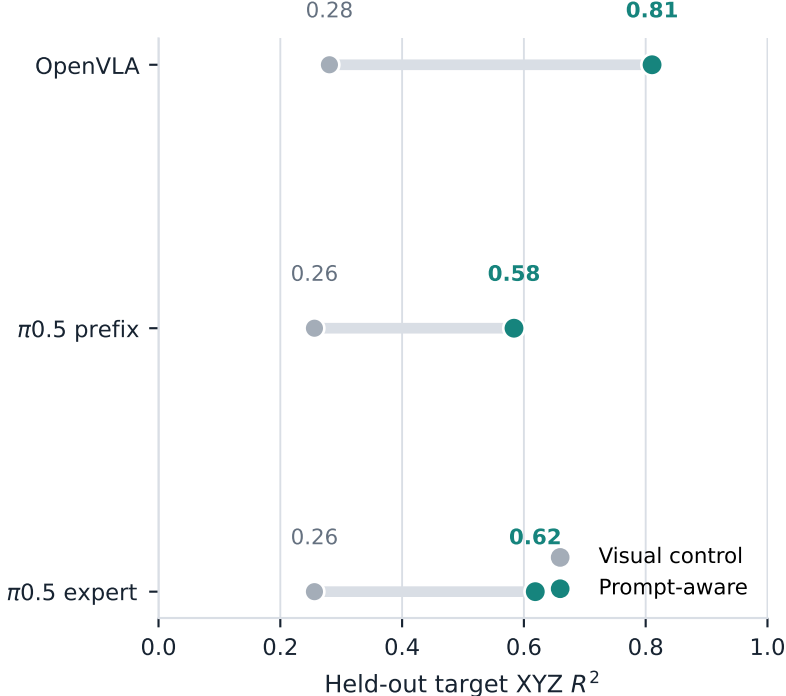
Goal-Centric World State in VLA Models: Multi-Pair Confirmation and Causal Boundaries

4 object pairs | 8 tasks | 400 rollouts | 15,275 states | 64 locked test episodes | 3 probe seeds

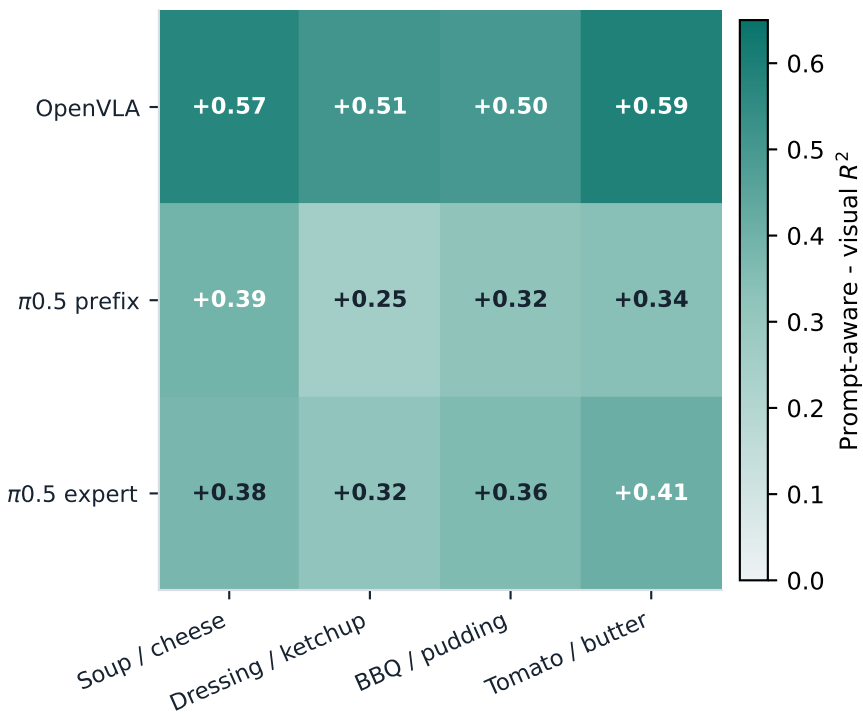
a Othello-style coordinate advantage



b Goal state survives the 7.9x scale-up



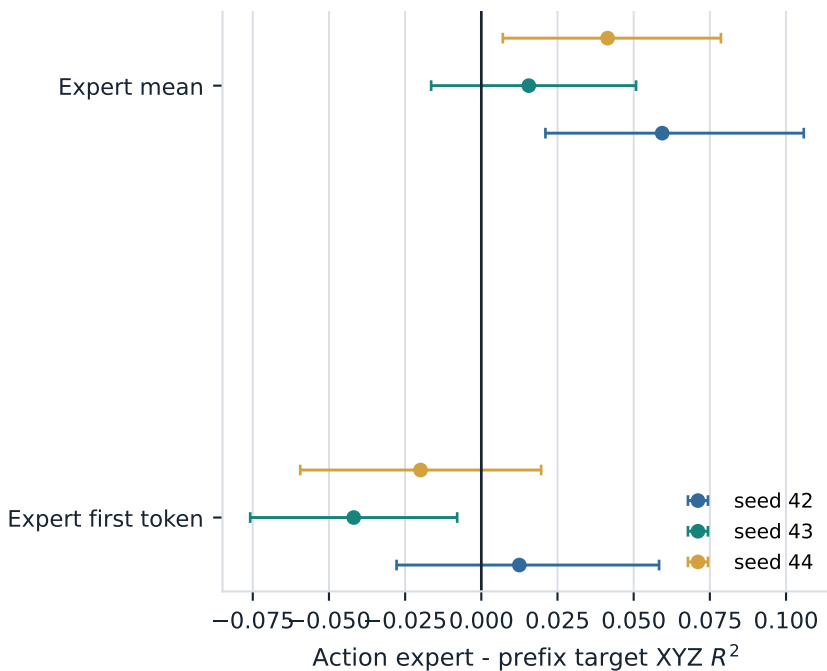
c Pooled gain is positive for every pair



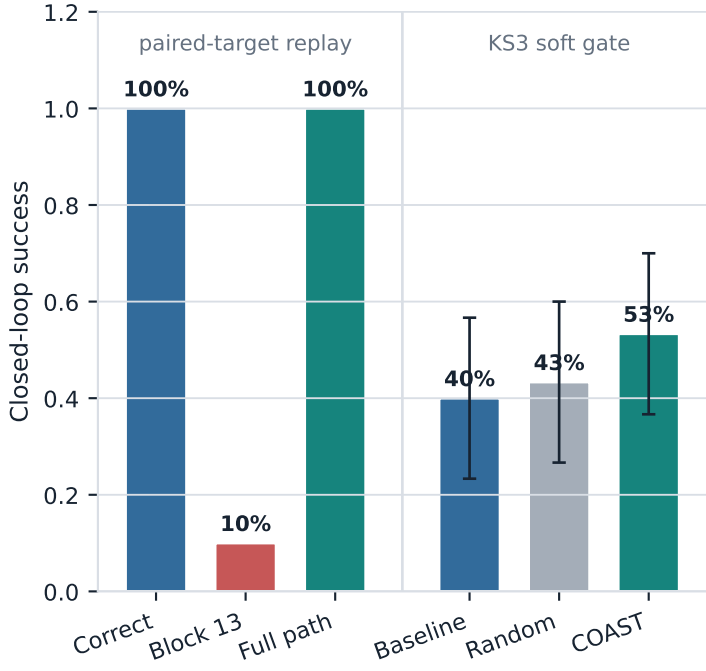
d Leave-one-pair-out transfer



e Propagation is real; extra decodability is modest



f Causal use depends on pathway-compatible operators



Panel f combines two preregistered causal tests; COAST uses a different task and checkpoint.